



Self-Aware Cognition

The agent need to know **the state of itself**, e.g. current position and flight behavior related to a landmark.

Landmark-relative Position Reasoning

Based on landmark's reference view, what is your current position relative to the landmark?
Return the option and the landmark's BBox.

- [A] Right [B] Front-Left
[C] Back-Right [D] Left

Landmark's BBox: [0.420, 0.361, 0.613, 0.687]



Future Observation Prediction

You can see landmark's reference view. Which candidate image is the resulting future view after taking a 90-degree clockwise?



Landmark's BBox: [0.420, 0.361, 0.613, 0.687]

Flight Behavior Recognition and Temporal Localization

Which of the following actions have you ever adopted in this first-person flight video?

- [A] Moving foward to the building [B] Square-shaped orbiting
[C] Move forward to the bridge [D] Square-shaped inspection

Time interval: A [1.00, 4.50]; B [4.50, 22.90]; D [1.00, 27.50]



Knowing the Self, Understanding the World

✿ **Focusing on dual-cognition capability required for UAV Agents**

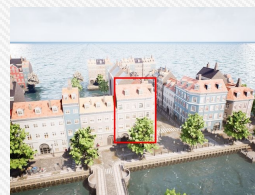
UAV-DualCog

✿ **New benchmark for dual-cognition evaluation in both image and video tasks.**

Tasks are generated based on a landmark, such as a waterfront building for example (see below) .

IMAGE TASK:

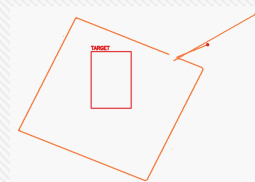
Tasks based on images captured from multi angles around the landmark.
Right is its front view.



(captured in scene)

VIDEO TASK:

Tasks based on first-person flight video of a trajectory around the landmark. Right is video's panonama view of a square-shaped inspection.



(rendered in pointcloud)

IMAGE TASK

VIDEO TASK



Environment-Aware Cognition

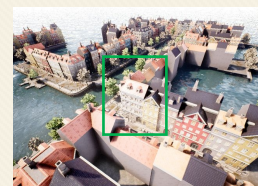
The agent need to know **the state of environment**, e.g. a landmark's position and visible time during its flight.

Self-relative Position Reasoning

Based on landmark's reference view, what is the landmark's current position related to you?
Return the option and the landmark's BBox.

- [A] Left [B] Front-Left
[C] Back-Right [D] Dront

Landmark's BBox: [0.369, 0.293, 0.630, 0.740]

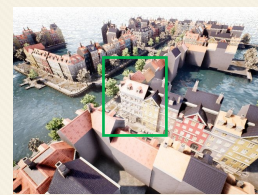


Future Observation Prediction

Based on landmark's reference view, which direction should you fly to get closer to it?
Return the option and the landmark's BBox.

- [A] Left [B] Front-Left
[C] Back-Right [D] Dront

Landmark's BBox: [0.369, 0.293, 0.630, 0.740]



Landmark Visibility Counting and Interval Reasoning

The reference image shows light gray mid-rise building with red roof and multiple windows. How many times does this landmark appear in your first-person flight video?

Count: 2; Time interval: [1.10, 4.50], [4.50, 22.90]

IMAGE TASK

VIDEO TASK